

Module 12

Budgeting

Learning Objectives

By the end of this module you should understand:

1. The role played by budgets in an organization
2. The problems surrounding budgets and how to solve these problems
3. The techniques of preparing a master budget and supporting schedules
4. The nature of discretionary expenditure and how zero-base budgeting can assist in allocating resources

Agenda

12.1 Introduction

12.2 Why Bother with Budgets?

12.3 Why Budgeting Gets a Bad Name

12.4 Budgeting in Action: The Go-Straight Trolley Company

12.5 Discretionary Expenditure and Zero-Base Budgeting

Review Questions

Case Studies

In module 9, we've covered the behavior of different types of cost; where costs are broken down into variable and fixed costs

Variable Costs

Costs that change in magnitude with production or sales levels

Fixed Costs

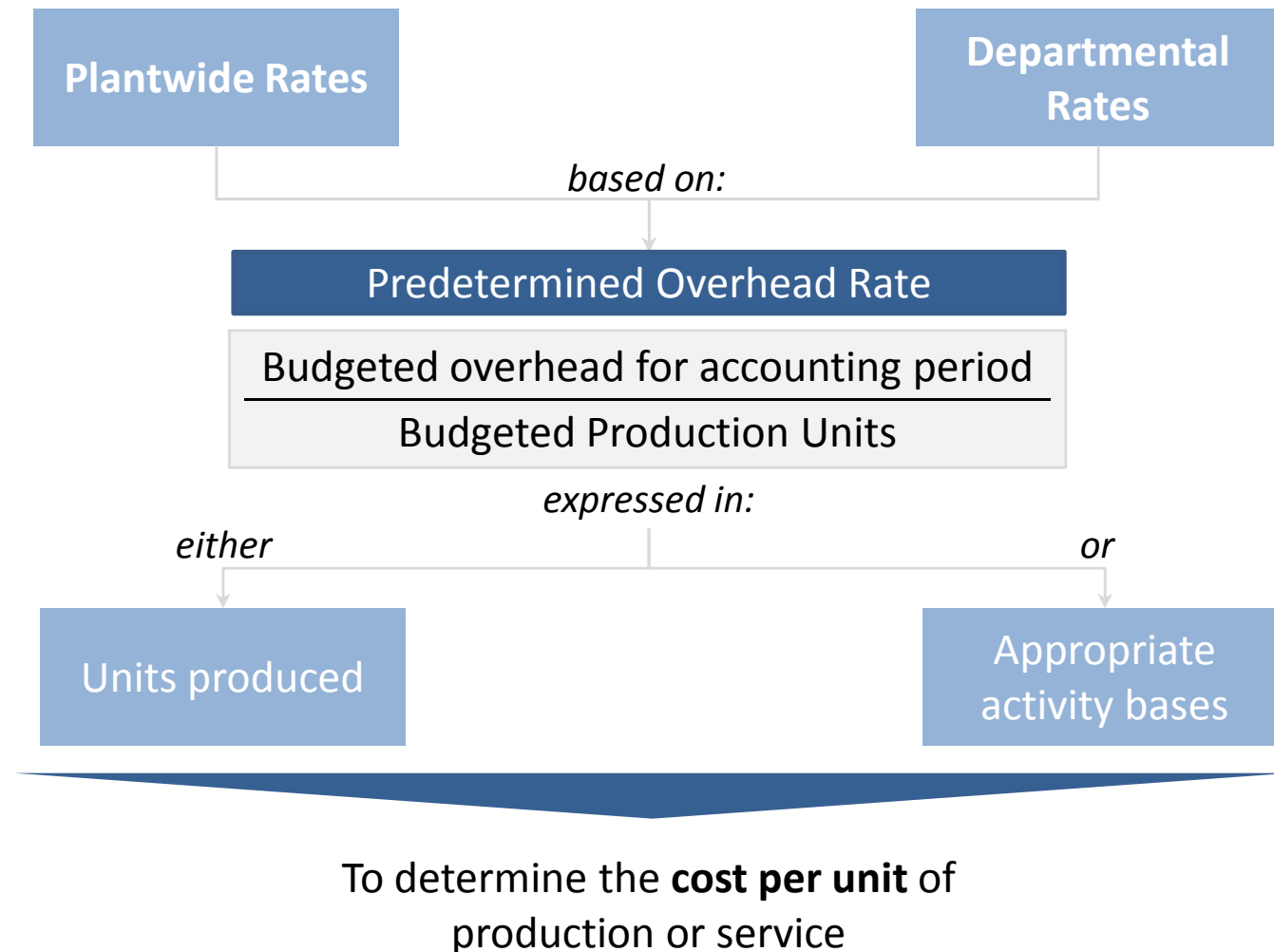
Costs that do **NOT** change as production or sales move up or down

provide:

Useful Managerial
Information

e.g. break-even points & the impact on profit of changing sales mixes

Module 10 described various procedures for spreading those costs using plantwide and departmental overhead rates



We've also seen in Module 11, the requirement to identify only relevant costs for decision making

Little space has been devoted so far to the predictive step in the management process "Budgeting"; which will be the subject of this module and the next

Budgeting

- A plan of action, usually expressed in numbers and amounts, which sets targets for individuals and for the business over a defined future timeframe
- Management must give thought to the future in order to plan to meet its threats and opportunities

Module 12

Sets out the procedures for constructing the budget

Module 13

Considers the mechanism available to management to monitor the actual performance against the plan

Agenda

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12.2 Why Bother with Budgets?

12.3 Why Budgeting Gets a Bad Name

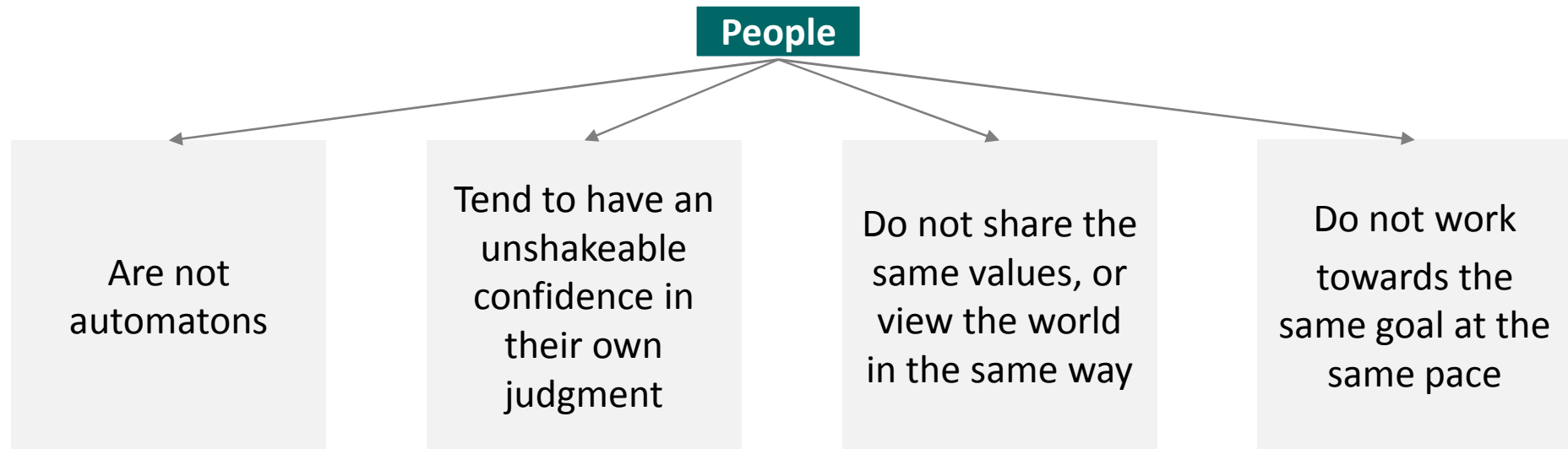
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Case Studies

We have to always remember the human factor in businesses; business is made up of people



Budgeting process should help in harnessing, directing and controlling employees' efforts and motivation

Budgeting has 4 attributes: Coordination, Planning, Motivation and Control

Coordination

It coordinates activities within the same organization



The budget will identify whether the aspirations of the sales team can be realized by the production department or whether these should be trimmed in order to match the limited output resources of production

Planning

It encourages planning



By planning, a megalomaniac personality of one senior official is countered by the marketplace. E.g.: a proposed twofold increase in the marketing budget for next year may not be sanctioned if sales targets are under pressure due to overseas competition or perceived quality

Motivation

It promotes motivation



Once the business's budget is broken down into its constituent parts, individual departmental heads obtain a clear picture of the route ahead for the next period, usually a year

Control

It provides a control benchmark against which to measure actual performance



Without a budget managers do not know how well their departments have done; they think they are performing satisfactorily but have no way of confirming this

issues

A budget is only a numerical expression of a plan, it is not the plan itself – referred to as strategic planning; which must precede the budget exercise

Strategic Planning

Strategic planning is the managerial exercise which attempts to ensure that a business accomplishes a sufficient process of innovation and change by:

- Allocating scarce resources (cash, critical management talent and technological know-how)
- Adapting to environmental opportunities and threats
- Coordinating activities so as to reflect the business's strengths and weaknesses

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Case Studies

There are some reasons why budgeting is getting a bad name; these should be overcome if the business is to use budgets as a significant management tool

Few of the reasons which must be overcome:

- 1 Time Taken
- 2 Lack of Top Management Commitment
- 3 A Form of Punishment
- 4 Responsibilities Are Blurred
- 5 Moving Goalposts
- 6 Budgeting Rewards Inefficiency

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The budget process should start as late as possible in the current year consistent with getting agreement by the start of the new year

In Big Companies

When to start
Budgeting?

Headquarters often start the budgeting process before the current financial year is six months old

Why so early?

A good budget process is an **iterative** one where first thoughts need to be amended by other managers' first thoughts

Why end up late?

Face-to-face meetings

Budget conferences

Because managers do not know how the current year is panning out (i.e. how accurate last year's budget is proving to be) before being asked for an assessment of next year

If this year is only a matter of a few months old, managers will disregard the early steps in the process and supply unsubstantiated guesses which will require amendment as the year progresses, thereby undoing the advantage of an early start

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Managers and departmental heads in the operating divisions very often resent the effort and self-control required of them as individuals to 'keep within the budget'

Reasons Behind this:

- 1** They see apparent **extravagant spending** on the part of senior management personnel
- 2** They suspect that this level of spending is not required to carry out the functions of the individuals involved
- 3** They suspect the spending continues to be incurred even after the budgeted resources given over to that activity have been consumed

A less-than-professional approach to their own responsibilities by top management

 *will ultimately lead to:*

A breakdown of the budget process where managers do whatever pleases them, disregarding previously agreed targets and limits

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Budgets are seen to be devices for control purposes only and for screwing down activities to the absolute minimum amount of resources required to do the job

Reasons behind this belief:

All discretionary expenditure such as entertaining and travel are stripped out

Surplus staff are removed and the remainder required to work in sweat-shop conditions

If this perception of budgets is widespread



Top management deserves to be removed from office

Through consulting with those employees who will be required to implement the budget, top management can guarantee support of those managers who had a hand in the construction of the budget

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Managers in operating divisions feel distressed when corporate or centrally incurred costs, over which they have no control whatsoever, are allocated to them

Examples of such costs:

Grouping wide marketing & promotional activities

Grouping executive salaries

Grouping expenses such as the corporate fleet of aircraft

Grouping expenses such as centrally located R&D laboratories

Corporate expenditure should be allocated to a budget holder who can be held responsible as it is unfair and demotivating to assess their performance on managing uncontrollable costs

Even among peer-level colleagues responsibility for costs can sometimes be blurred – consider the following problem

Example

- A multinational food company based in central Europe found itself with a temporary surplus of nutritional scientists in the R&D labs
- This surplus was due to a temporary downturn in a specific sector of the market.
- In an effort to retain their services (lest they be poached by a competitor to whom they might reveal recipes and formulae for new products) the company transferred these scientists to the tasting department of a nearby production plant.
- The supervisor of the tasting department welcomed the scientists for the first month until he read from his cost report that his budget for temporary tasters had been exceeded by three times due to the salaries of the R&D scientists.
- 'I don't give a hoot for company policy,' he protested.
- 'If I can hire tasters at 100 Francs a day only when needed, why should I be stuck with 300 Francs each day per scientist whether they're needed as tasters or not? The R&D lab can take back the difference between this month's charge and what I would have paid out in normal circumstances.'

This scenario is typical of situations where a budgeting system attempts to reflect managerial responsibilities

Several scenarios pose themselves in such a situation... which shows how budgeting responsibilities can be sometimes blurred

The accountant who authorized this cross-charge may agree that only 100 Francs per active hour be charged to the tasting department and that the difference be charged to a central account (e.g. **Spare scientific capacity**) since the problem has been caused by a top-level decision to retain scarce scientific expertise.

The supervisor of the tasting department may even argue that the scientists are unreliable as tasters and that if he had authority in **keeping with his budget** responsibility he would get rid of the scientists and hire his part-timers who were experienced in tasting

Supervisor does not have the authority to discharge or transfer the scientists that he might ordinarily possess over his regular staff. Therefore the 'Spare scientific capacity' account balance should be **charged to the senior executive** who took the decision to retain the scientific manpower

The company should not term this surplus account an **asset**, an argument which could be mounted around the fact that the decision to keep scarce skills implies a future cost savings in hiring and training new recruits when business turns up – which wouldn't be accepted by the auditors

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Another complaint often leveled by managers is that circumstances can change early in the year that can invalidate the numbers which were so carefully budgeted

If their numbers are not updated in the light of the
changed circumstances

*e.g. the collapse of a major customer
or supplier*



The resultant comparisons with actual
performance are meaningless

At the same time if the budget is adjusted to reflect irrelevant factors; the initial effort which was invested in the preparation of the original figures is seen to be misplaced

Companies can avoid these managerial misgivings in one of two ways –both are meant to maintain the integrity of the budget process

Management can adopt:

Rigid view towards the original budget

The original numbers remain the benchmark against which subsequent performance will be measured regardless what happens next month

The accounting staff strips out the variance caused by the external event before highlighting those variances which can be viewed as the responsibility of the managers involved



The managers' performance is assessed on the up-to-date expectation but the company does not lose sight of the original view of the future as represented by the beginning-of-year budget

A rolling budget

As the current month expires, another month is added on to the budget 12 months away



The budget is always a 12-months' view of the future and can be amended as soon as the 'disturbing' events become known

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“Budgeting rewards inefficiency” represents the notion that costs and revenues are very often budgeted to increase by a constant percentage

Example:

Budget controller to production director:

‘We want to boost sales by 10% next year. Work out your production budget accordingly and let us know what extra resources you will need.’

Chief executive to marketing manager:

‘If we were to increase your budget by 10% next year, what would you do with the extra resources – people or promotions?’



This is a common approach in the private sector, particularly in companies which have had a track record of satisfactory profits

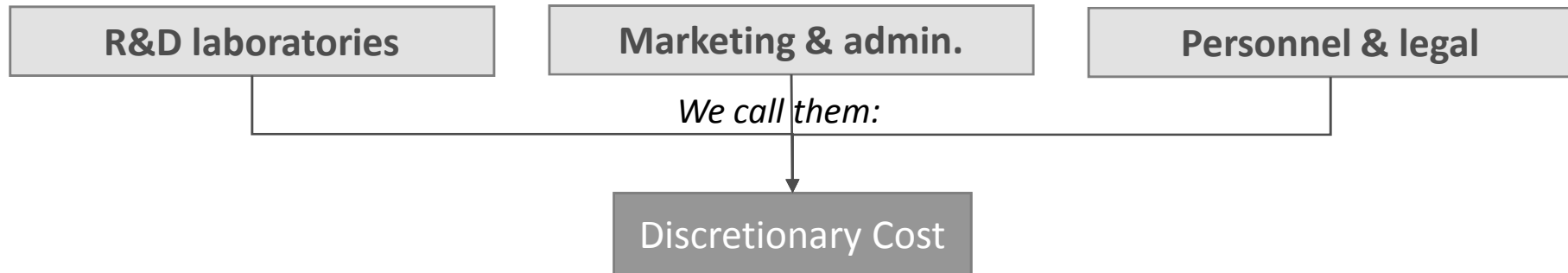
This approach to budget uplifts irritates many managers who regard it as a technique for rewarding inefficiency

Two Scenarios:

	Will be given an additional percentage uplift	Will be given a downward percentage cuts
Inefficient Departments	More corporate resources will be wasted by these departments	Only asked to cope with a reduction in resources which may not even remove all the inefficiencies
Efficient Departments	The tightly run, well-managed departments are penalized accordingly	Must make substantive cuts which will have a negative effect on their ability to deliver a quality service

Driving out inefficiencies from budgets is one of the most difficult managerial tasks, particularly in areas where the benchmarks of performance are blurred

Examples:



Example

A company making DVD cabinets knows how much to expect by way of material costs, labor costs and energy costs but it does not really know how much it should spend on its personnel department.

Top management should encourage competition for surplus resources within the organization while, at the same time, not recoiling from taking allocation decisions even if they are based on intuition

Remember: Discretionary costs are those costs that can be influenced by managerial decision

Engineered Costs

Costs engineered into the product

Should be incurred every accounting period

e.g.: the cost of chipboard veneer, metal trim and direct labor or certain critical items of manufacturing overhead

Discretionary Costs

Costs that managers can reduce relatively quickly

Need not be incurred every accounting period

e.g.: machine maintenance, administrative support and R&D

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Case Studies

Instead of setting out the steps for the preparation of a budget in a sequential and theoretical manner removed from any commercial context, let us use this case

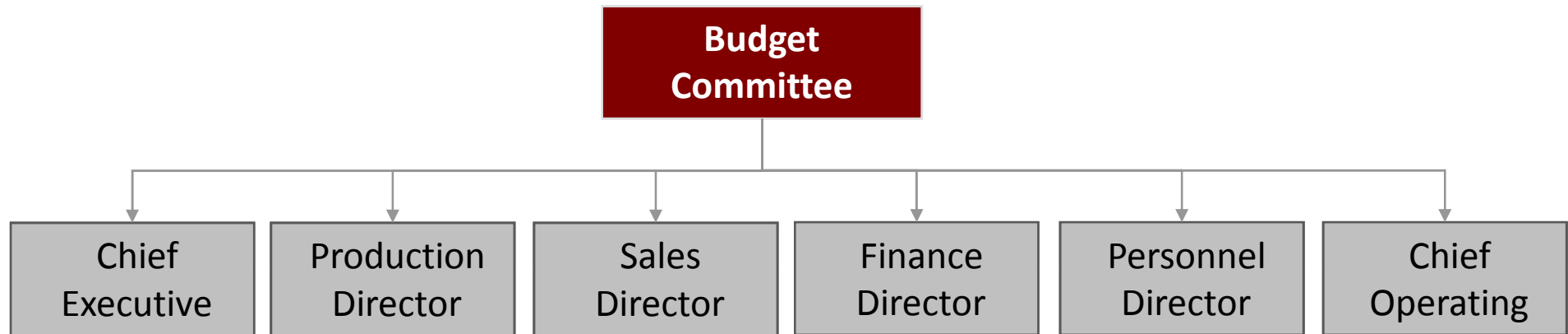
Case Study:

- The Go-Straight Trolley Company is a successful company which has completed its financial year to 31 March 20x3 and is planning for the next 12 months. Clearly the exercise would commence several months before the year-end, here it would be a few days at the end of March 20x3.
- It manufactures and sells a much sought-after product, a supermarket trolley that can be pushed by the customer without effort and in a straight line. The special wheels and bearings which permit such easy motion are the result of careful research over the past three years of the company's trading history. The management of the company is keen to grow slowly without overreaching the company's resources; the managers have read too many stories of similar sized companies expanding too quickly with borrowed money only to go into liquidation when the market turned against them, and they are determined to avoid such a fate.

Setting the budget commences several months before the year-end

The chief executive would issue general guidelines to the principal heads of departments, while in larger organizations a more formal budget committee is convened

In Large Organizations:



The financial controller would probably be appointed **budget officer** whose task would be to drive the exercise forward to final approval by the budget committee on behalf of the BOD

The chief executive masterminds the whole budgeting process, through the report and the guidelines he gives

The guidelines would contain:

- A commentary by the chief executive on the year just finishing
- Reasons for the superior (or inferior) performance against budget
- His view on the potential for growth
- A report on the expected product launches in the forthcoming year
- The relevant economic indicators to the business (including indicators of consumer spending
- Rate of construction of supermarkets and shopping centers and family expenditure patterns)
- View on growth.

He or she is the best placed person to weigh the potential for growth in output against the company's ability to support this growth from its own resources or from borrowings

The Chief Executive's report may look like this:

Memorandum from Chief Executive:

Memorandum from Chief Executive

To all Functional Managers: Budget Preparation for year to 31 March 20x4

As a background to the forthcoming budget process I should like to sketch in a few indicators which I should like you all to keep in mind in the next few weeks as we work towards closure on the budget for year ending 31.3.04.

Case Study (Continued)

Memorandum from Chief Executive (Continued):

The final unaudited figures for the year ended 31.3.03 will reveal a healthy profit and a strong balance sheet which is reproduced below:

Go-Straight Trolley Company: Balance sheet as at 31 March 20x3

Fixed assets	€	€
Plant and equipment	75 000	
Less: Accumulated depreciation	<u>25 000</u>	50 000
Current assets		
Inventories:		
Metal: 600 metres @ €1	600	
Finished trolleys: 480 @ €55	<u>26 400</u>	
	27 000	
Debtors	7 500	
Cash	<u>17 500</u>	52 000
Total assets		<u>102 000</u>
Owner's equity		
Share capital	30 000	
Retained earnings	<u>69 500</u>	99 500
Creditors		<u>2 500</u>
		<u>102 000</u>

Case Study (Continued)

Memorandum from Chief Executive (Continued):

This performance is a tribute to our carefully targeted growth in specialised sectors of the market, together with the ongoing good work of our R&D engineers in keeping our product technologically superior.

But there is still a reluctance on the part of many supermarket chains to buy the product due to its relatively high cost of €65. This is some 30 per cent higher than our nearest competitor and although they recognise the advantages of trolleys which move without steering effort they nevertheless are prepared to purchase inferior products. Eventually consumer pressure will make them change over to our product but I do not want us to contemplate a drop in either price or quality. Quite the reverse, I want us to consider a modest increase in price in order that we retain the image of quality and service which sets us apart from the competition.

I know many of you want to expand production so that we can attack other markets such as airports, coach stations and rail stations. We have commissioned a marketing study on such possibilities which will not be available before September. I therefore want you to consider only a modest increase in units sold this year. It is far better that we build a secure platform for growth both in terms of market share and financial soundness rather than charge ahead for new sales just to improve our turnover ratios.

Case Study (Continued)

Memorandum from Chief Executive (Continued):

The economic indicators which I have studied indicate a slowing down in consumer spending next year. The continued high interest rates are having an effect on disposable incomes; therefore we can expect fewer orders from the electrical stores and furnishing outlets. A modest growth in household spending stores (grocery stores and the like) can be expected. And the much heralded developments in UK megacentres for shopping have not yet materialised. We can expect some movement in 20x5.

One final word: I have detected some resentment among some of you about the high level of spending on R&D engineering. Those who harbour such feelings should remember that our company's history is based on a unique design of wheel and bearing. Had it not been for our early investment in R&D we would not be working for this company today. We must keep ourselves technologically in the forefront. The chief engineer is currently working on a device which will save the customer and check-out operator from having to unload and reload the trolley at the check-out. The designs are still in their infancy but we anticipate significant demand for this if we can be first-to-market. But this design and prototype stage costs money. So I want to allocate 3 per cent of sales revenue to R&D. Those of you who grumble at such expense should contemplate the future for this company when our competitors catch up with us and we have nothing new to offer the market.

Please let the financial controller have your first thoughts on next year's budget by the end of the week. We can then have a meeting to iron out different views and go firm on the numbers by 1 April.

Case Study (Continued) –The sales team would be called together by the Sales Director to plan their numbers for the following year

Sales
Director

‘You have all received a copy of the boss’s guidelines. I’m a little disappointed at the request to raise prices because even at €65 some of our principal customers were complaining.’

Sales
District
Manager

‘But not all of them! Qualityrama expressed surprise to me that in their latest order, trolleys were invoiced at €65. As we had held our prices stable for nine months they were expecting – and I suggest would not have complained at – a price rise. €70 per trolley would be the figure which I think could be accepted by our customers. This would be a 7.6 per cent rise against Bonecrusher’s 12 per cent rise and Wibbly Wobbly’s 10 per cent rise.’

Sales
Director

‘OK, €70 seems reasonable but we may have to be prepared to discount this to some of our larger customers who show reluctance. But I don’t think we should flag this up in the budget because it is not going to be a big amount. Now what about the quarter-by-quarter sales profile?’

Sales
Executive

‘I have done some preliminary figures at €70 per trolley which have been based on this year’s profile and I’ve allowed for a 5 per cent increase in sales units due to the impact of our new advertising material. Here are my numbers:

Case Study (Continued) –The sales team would be called together by the Sales Director to plan their numbers for the following year

Quarter	Estimated sales units
1. to 30 June 20x3	2 500
2. to 30 September 20x3	2 800
3. to 31 December 20x3	1 700
4. to 31 March 20x4	3 000
5. to 30 June 20x4	3 000

‘The anticipated drop in units in Quarter 3 follows the trend in previous years. As our customers are stocking up for Christmas their cash flow prevents any significant outlay on such items as trolleys. But after Christmas we can expect a replacement program to be embarked on and for this to be carried forward, together with new business, into the first quarter of next year.’

Advertising Manager

‘I really think we want to step up the spend on advertising this year. I agree that the current campaign is having an effect although we were taking a risk in portraying a housewife being pinned against a freezer by one of our competitor’s trolleys. I should like to argue for similar funding to R&D, say, a 5% spend linked to sales revenue in addition to the €3,000 per quarter fixed salaries.’

The production team would have a similar meeting

Production
Director

‘Given the chief executive’s view on not wanting to ramp up production next year in advance of sales I suspect we should examine the sales team’s view on units to be sold and cost our production activity on these.’

Production
Controller

‘A word of caution here. Our suppliers are still fairly erratic despite our best efforts to signal demand to them in plenty of time. I should like to argue for having raw material inventories on hand equivalent to 10% of next quarter’s requirement. And another thing, despite these apparently well thought out sales figures, the sales people are always wanting rush orders put through because they have messed up their forecasting. From the experience of this year I think we should have finished goods inventory equal to 20% of expected sales next quarter.’

Purchasing
Officer

‘The price of wire metal will go up midway through the year but I’ve negotiated a 12-month purchasing bulk discount with our principal suppliers. So the cost per meter will be €1 on average for next year.’

Production
Controller

‘The cost per hour for our skilled labor will increase marginally but I anticipate this will be offset by increased speed of operations due to the new equipment we purchased last year.’

All in all, the variable costs and fixed costs would look like this:

	per trolley
Direct material	€
10 metres of wire metal @ €1 per metre	10
Direct labour	
1.5 hours @ €20 per hour	30
Variable overhead (absorbed by direct labour hours)	
1.5 hours @ €10 per hour	<u>15</u>
Variable cost per trolley	<u>55</u>
Fixed manufacturing overhead	€2 400 per quarter
(includes depreciation on plant and equipment	€1 500 per quarter)
Consumables	€2 000 per quarter

The finance and administration team would also discuss the forthcoming year:

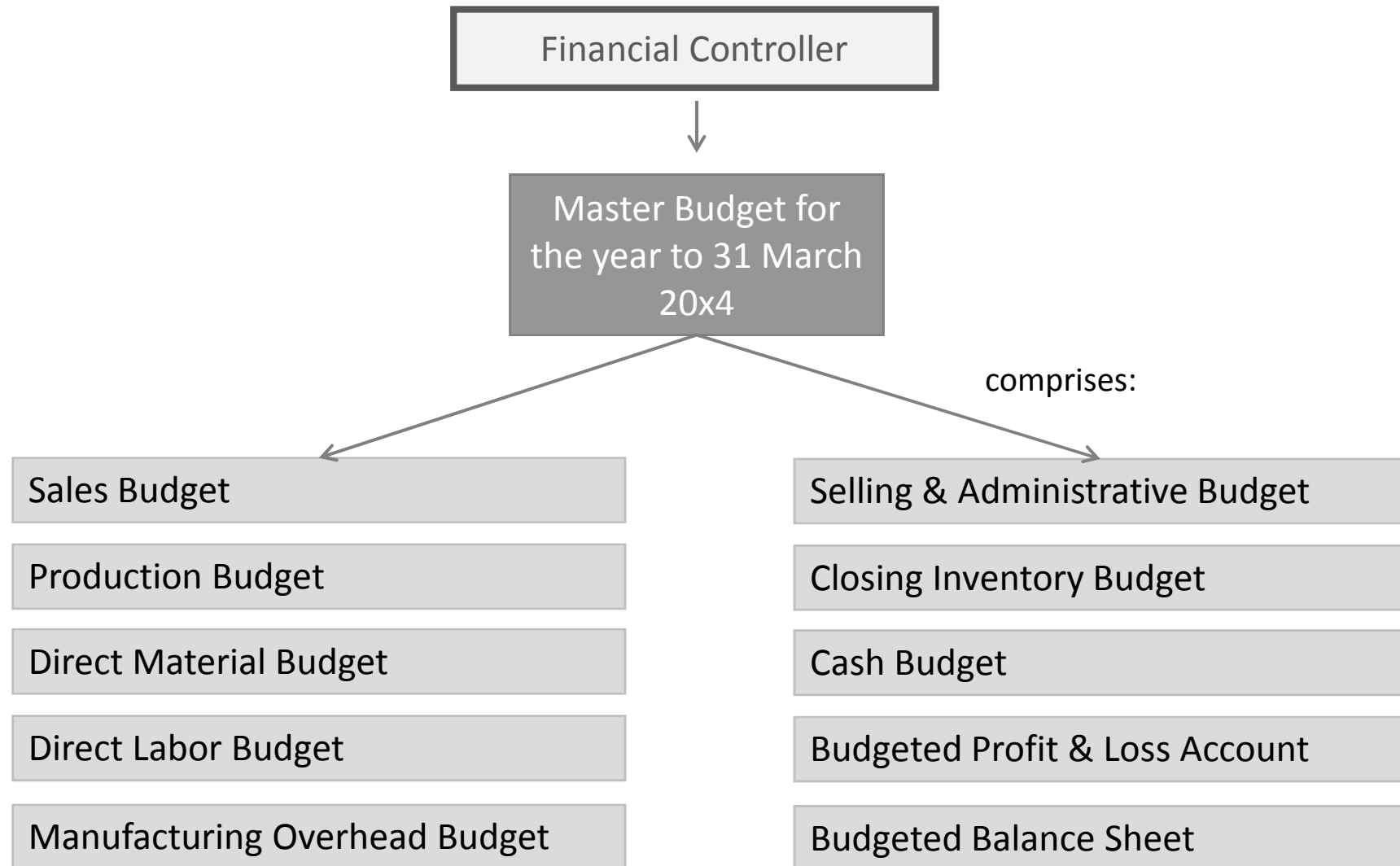
Financial
Controller

‘We must attempt to tighten up on debtors’ collection next year. We should plan for 70 per cent of sales being collected in the quarter of sales with the balance being collected in the next quarter. Yet again we had no bad debts this year due to the quality customers our sales team is targeting, so we should not predict any for next year.’

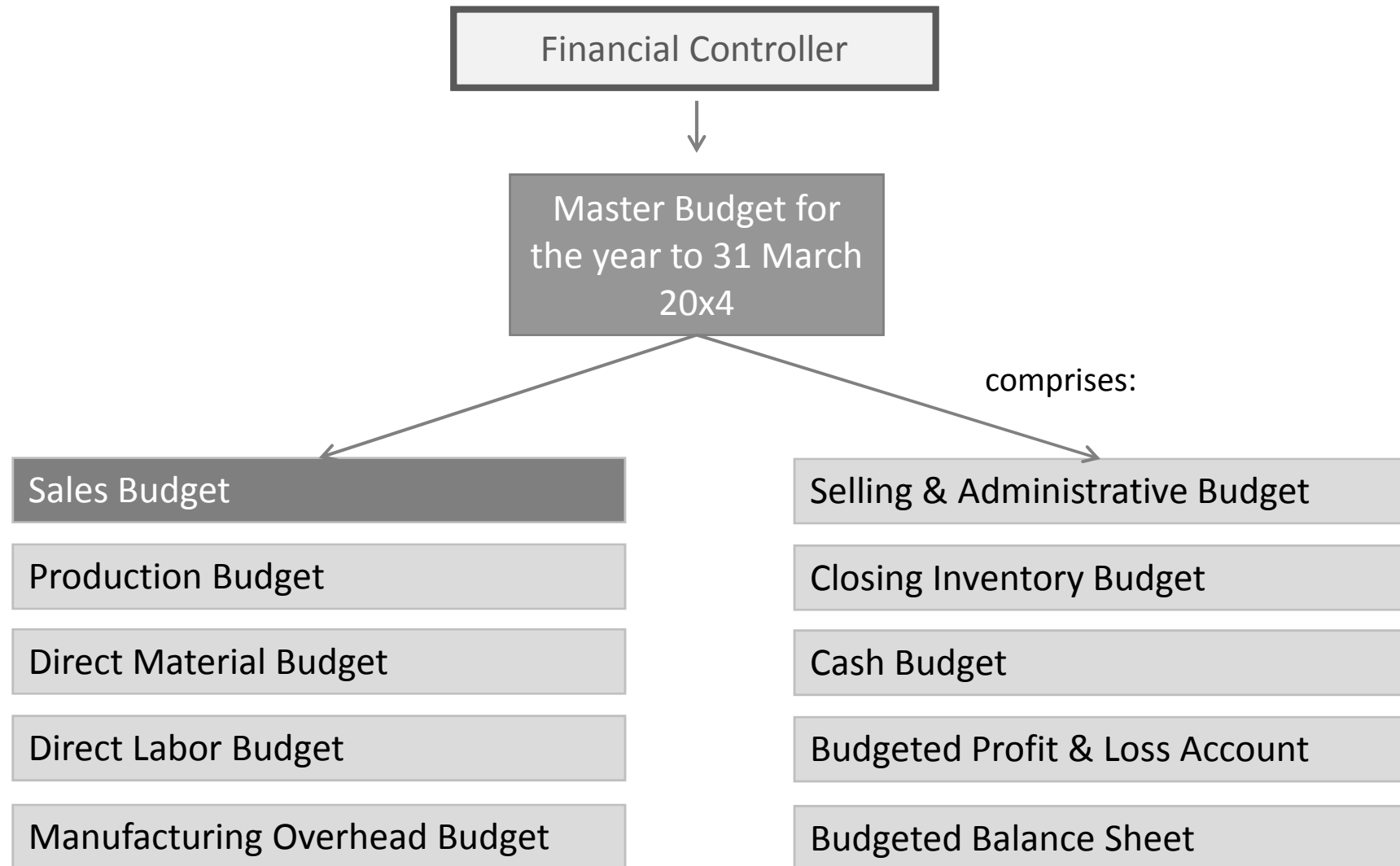
Financial
Director

‘We should be careful then about not paying our creditors too quickly. I suggest that raw material purchases should be paid for 60% in the quarter of purchase and 40% in the next quarter. All other selling, advertising and administration costs, including administrative salaries of €2,000, will be paid for, as usual, in the quarter they are incurred.’

These reports and supporting schedules would be submitted to the financial controller whose task would be to draw up the company's master budget



These reports and supporting schedules would be submitted to the financial controller whose task would be to draw up the company's master budget



Sales budget is the fulcrum on which the entire budget is balanced; as it tracks the profile of sales and converts these sales into cash receipts

12.4.1 Sales Budget for Year Ending 31 March 20x4 (Schedule 1)

(a)	Q1	Q2	Q3	Q4	Total
Turnover in units	2 500	2 800	1 700	3 000	10 000
Price	× €70	× €70	× €70	× €70	× €70
Recorded sales	€175 000	€196 000	€119 000	€210 000	€700 000
(b) Cash inflow from sales					
Last quarter's collection	€7 500	€52 500	€58 800	€35 700	€154 500
This quarter's collection	122 500	137 200	83 300	147 000	490 000
	€130 000	€189 700	€142 100	€182 700	€644 500

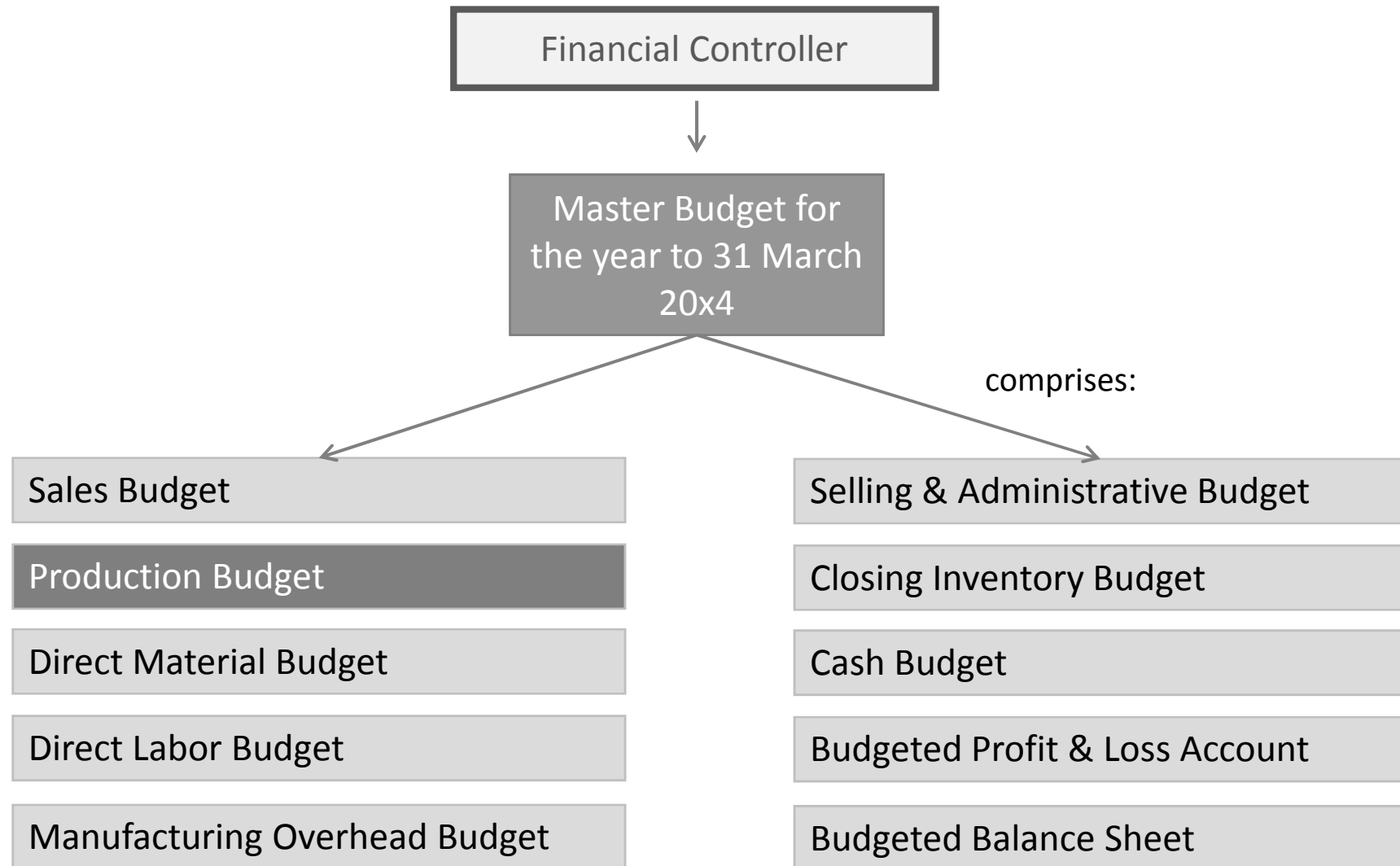
*70% of €175,500,
is to be collected in
Quarter 1*

*The rest €52,500
will be collected
next quarter*

*€147,000 in Quarter 4 &
€63,000 next year which
will be the debtors figure
for the budgeted balance
sheet at 31 March 20x4*

But the cash flow impact of these sales must be assessed, for cash is more important to a business day by day than is a profit figure

Master Budget



The sales targets of Schedule 1 would then be converted into production numbers

12.4.2 Production Budget for Year Ending 31 March 20x4 (Schedule 2)

	Q1	Q2	Q3	Q4	Total
Sales units	2 500	2 800	1 700	3 000	10 000
Add: Planned units in closing inventory	560	340	600	600	600
	3 060	3 140	2 300	3 600	10 600
Less: Units in opening inventory	480	560	340	600	480
Units required	2 580	2 580	1 960	3 000	10 120

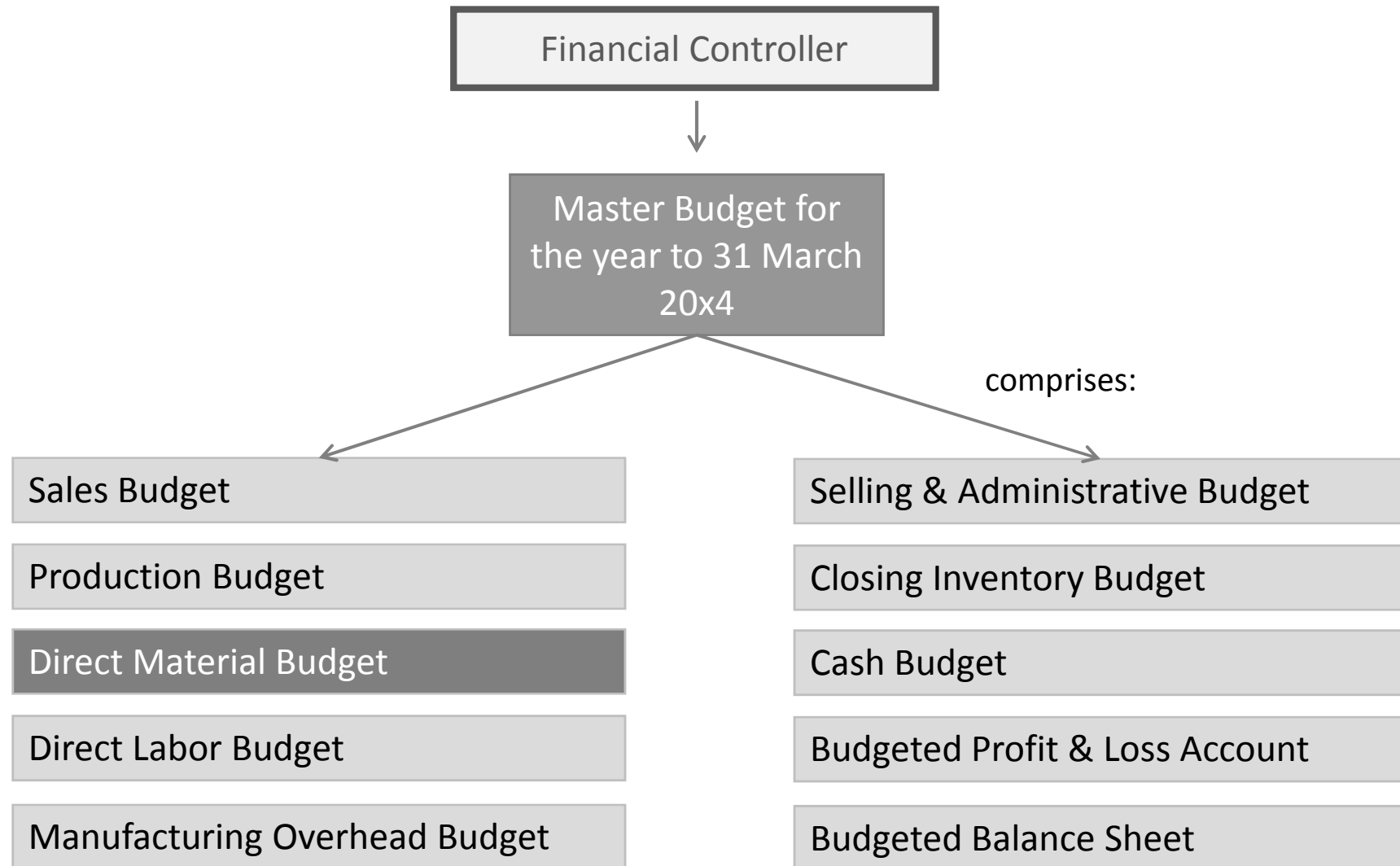
20% × Q2's 2,800 units

Therefore in Q1 3,060 units should be manufactured

But the opening balance of inventory of finished units Q1 480 would reduce this requirement

The production budget would take account of the production controller's observation that inventory of finished units should be carried to meet the uncertainties of demand

Master Budget



The production unit's budget must now be converted into a cost budget, comprising direct materials, direct labor and manufacturing overhead

Three Components to the direct materials budget

→ The consumption of metal

→ The purchases required to meet the production schedule

→ The payment to creditors profile

This schedule is self-explanatory, being based on the production schedule just calculated

12.4.3 Direct Materials Budget for Year Ending 31 March 20x4 (Schedule 3)

(a) Consumption of direct material

	Q1	Q2	Q3	Q4	Total
Units required	2 580	2 580	1 960	3 000	10 120
Direct materials metres of metal	× 10	× 10	× 10	× 10	× 10
Metres of metal required	25 800	25 800	19 600	30 000	101 200
Cost per metre	€1	€1	€1	€1	€1
Total cost of metal consumed	€25 800	€25 800	€19 600	€30 000	€101 200

This profile would allow for a rise in price to be recorded midway through the budget year if necessary

The starting point for the purchases budget is the number of meters of metal required to manufacture the units to be made (from Schedule 2)

(b) Purchases budget

Direct materials required (in metres)	25 800	25 800	19 600	30 000	101 200
Add: Planned closing inventory	2 580	1 960	3 000	3 000	3 000
	28 380	27 760	22 600	33 000	104 200
Less: Planned opening inventory	600	2 580	1 960	3 000	600
	27 780	25 180	20 640	30 000	103 600
Metres required					
Cost per metre	× €1	× €1	× €1	× €1	× €1
Total cost of metal purchased	€27 780	€25 180	€20 640	€30 000	€103 600

*Taken from the closing
balance sheet as at 31
March 20x3*

But again the production controller's wishes for inventory must be built in to the figures; where 10% of the next quarter's production must be added to this quarter's requirements

Note the total column in this cash outflow schedule bears a resemblance to the quarter-by-quarter profile in the purchase budget

(c) Cash outflow for purchases

	Q1	Q2	Q3	Q4	Total
Creditors end 20x3	€ 2 500				€ 2 500
Purchases:					
Q1	16 668	€11 112			27 780
Q2		15 108	€10 072		25 180
Q3			12 384	€ 8 256	20 640
Q4				18 000	18 000
	€19 168	€26 220	€22 456	€26 256	€94 100

The company will pay outstanding creditors from the end of last year, €2,500

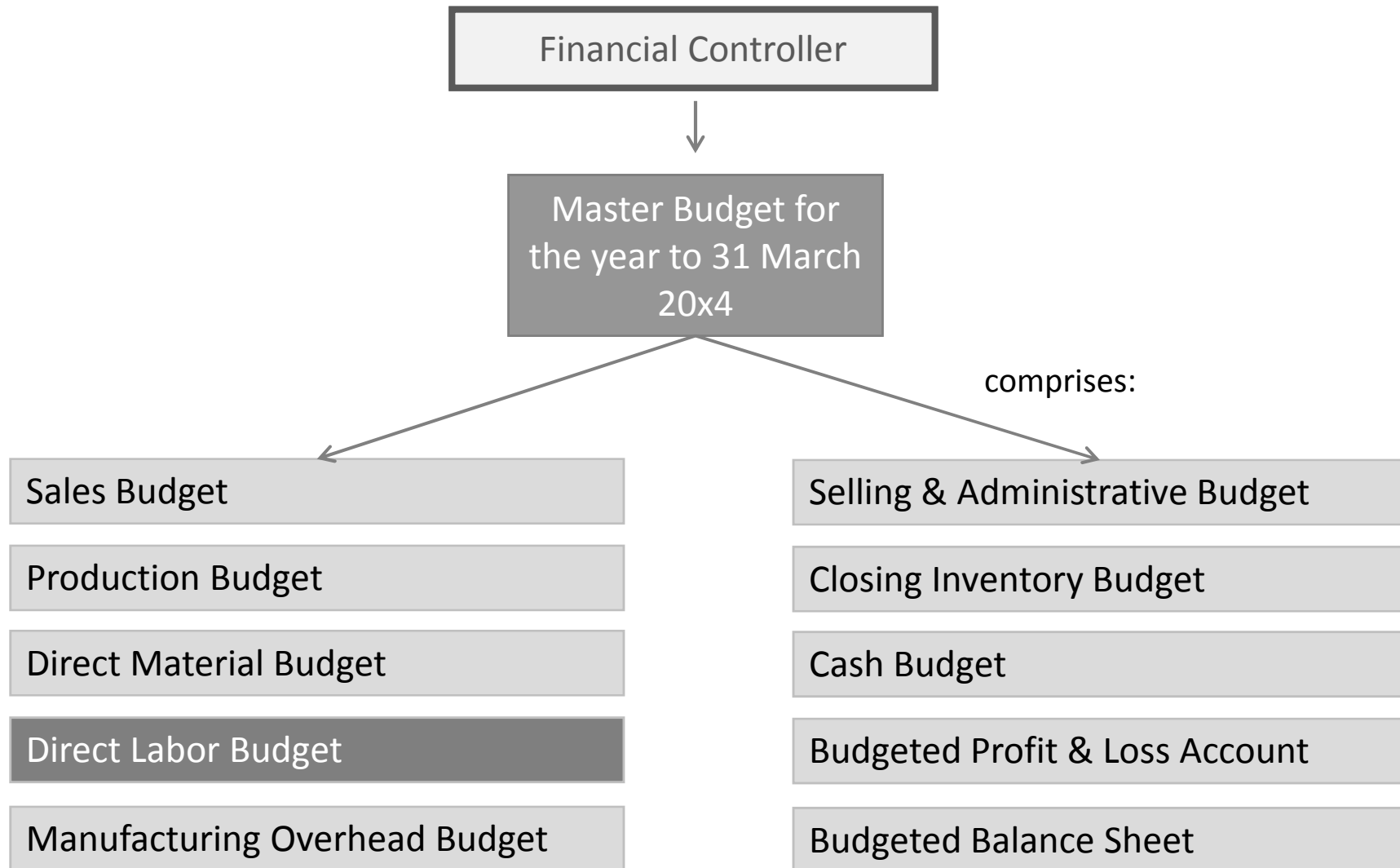
60% of €27,780 purchases made in Q1

The balance of Q1's purchases, €11,112, will be paid for in Q2

Q4's purchases of €30,000 will be paid for: €18,000 in Q4 and €12,000 in next year's Q1

The cash outflow schedule splits up the purchase budget into the quarters which match the cash payments

Master Budget



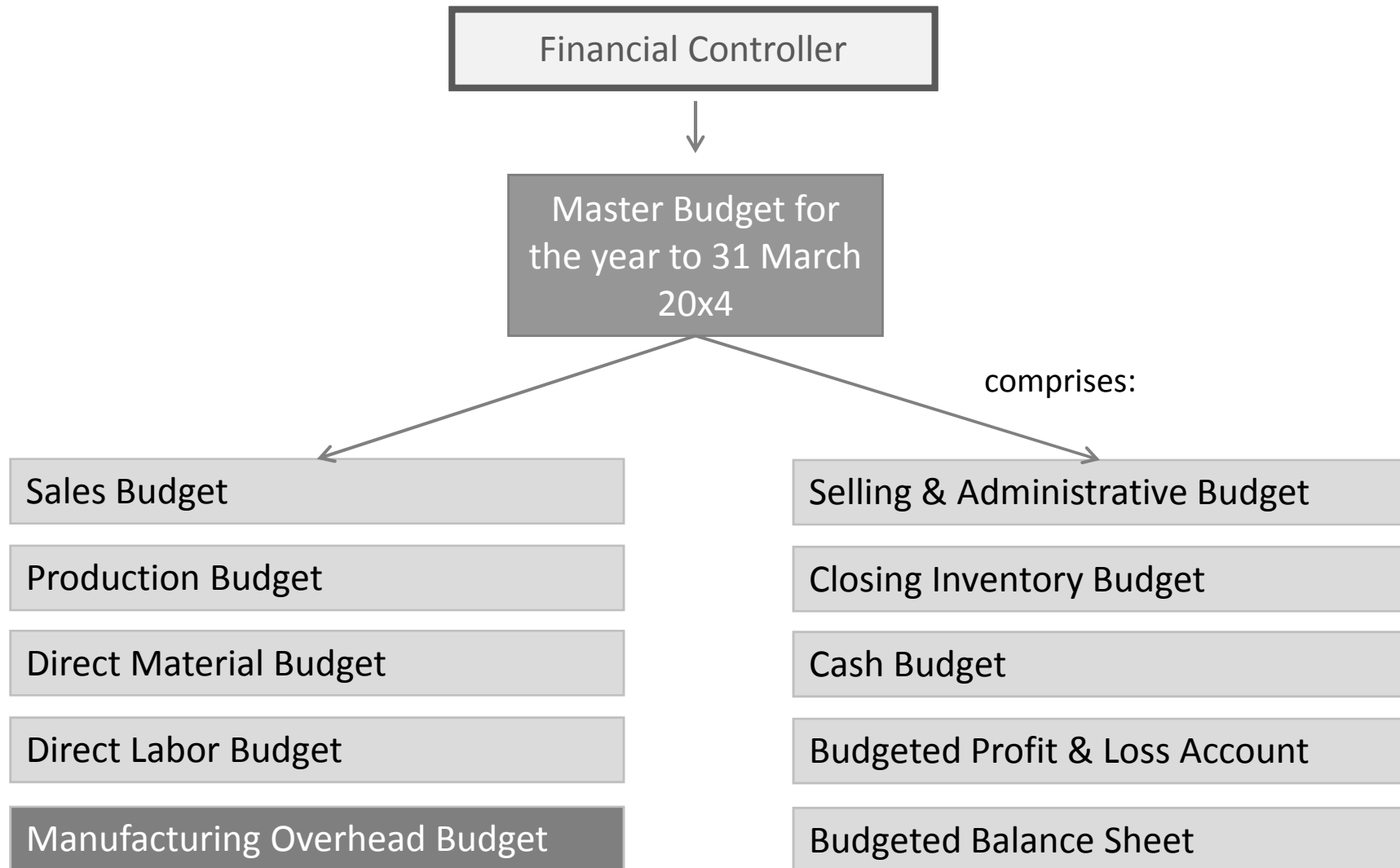
The direct labor budget presents no difficulties because labor cannot be ‘inventoried’; schedule 4 is based on the production units calculated in Schedule 2

12.4.4 Direct Labour Budget for Year Ending 31 March 20x4 (Schedule 4)

	Q1	Q2	Q3	Q4	Total
Units required	2 580	2 580	1 960	3 000	10 120
Hours per unit	× 1.5	× 1.5	× 1.5	× 1.5	× 1.5
Total no. of labour hours	3 870	3 870	2 940	4 500	15 180
Cost per hour	× €20	× €20	× €20	× €20	× €20
Total cost of direct labour	€77 400	€77 400	€58 800	€90 000	€303 600

The actual cash outlays for labor will match exactly the calculations of this schedule

Master Budget



The manufacturing overhead budget is equally straightforward except for the separation of variable & fixed costs and the deduction of depreciation

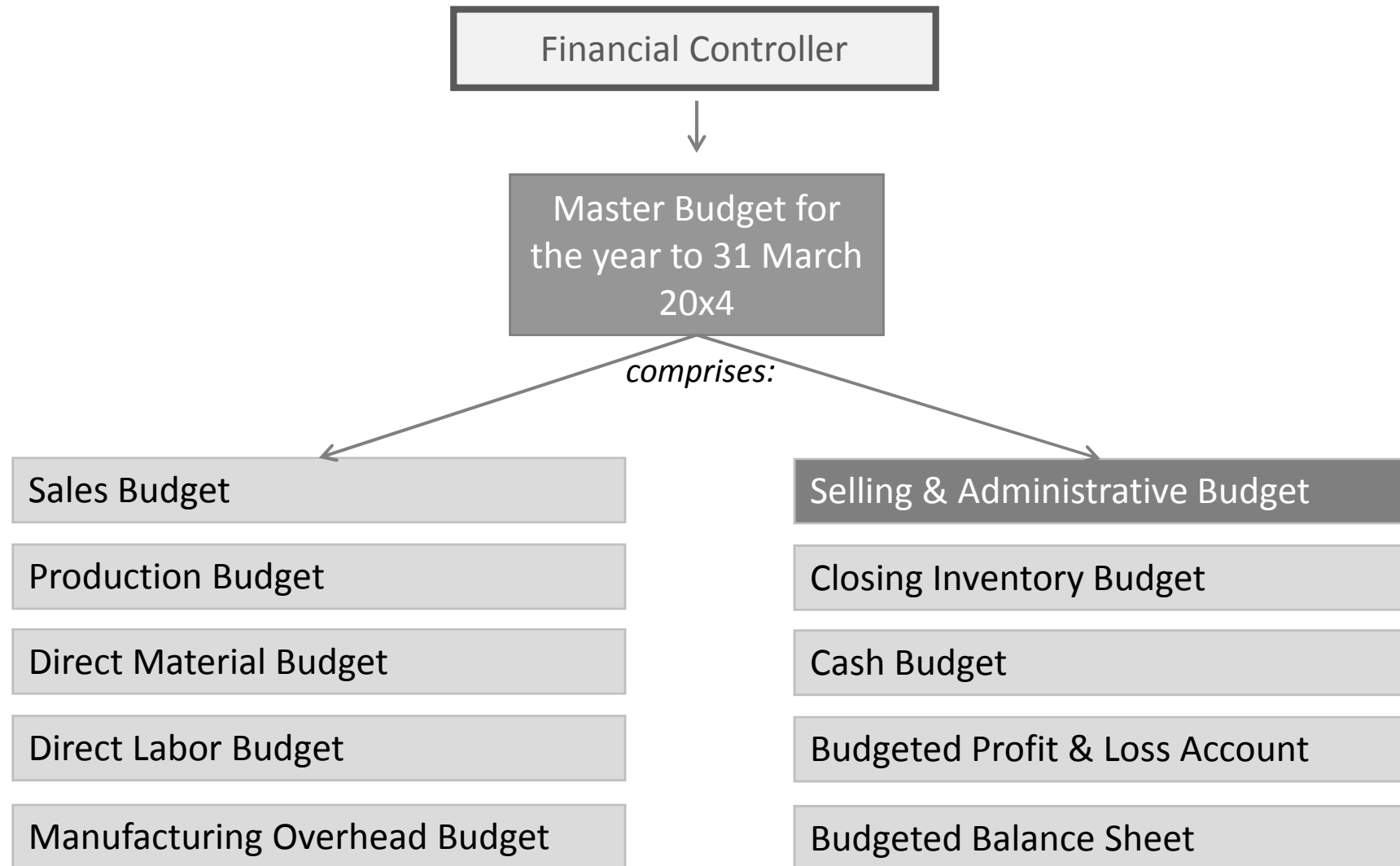
12.4.5 Manufacturing Overhead Budget for Year Ending 31 March 20x4 (Schedule 5)

	Q1	Q2	Q3	Q4	Total
Total cost of direct labour	€77 400	€77 400	€58 800	€90 000	€303 600
Absorption rate for variable overhead	€0.50	€0.50	€0.50	€0.50	€0.50
Variable overhead	€38 700	€38 700	€29 400	€45 000	€151 800
Fixed overhead	2 400	2 400	2 400	2 400	9 600
Total manufacturing overhead	€41 100	€41 100	€31 800	€47 400	€161 400
Less: Depreciation (non-cash)	1 500	1 500	1 500	1 500	6 000
	€39 600	€39 600	€30 300	€45 900	€155 400

*50% of direct labor cost (or
€10 per direct labor hour)
as mentioned by the
production controller*

Management would expect a detailed analysis of variable and fixed overhead and would not accept this budget without such an analysis

Master Budget



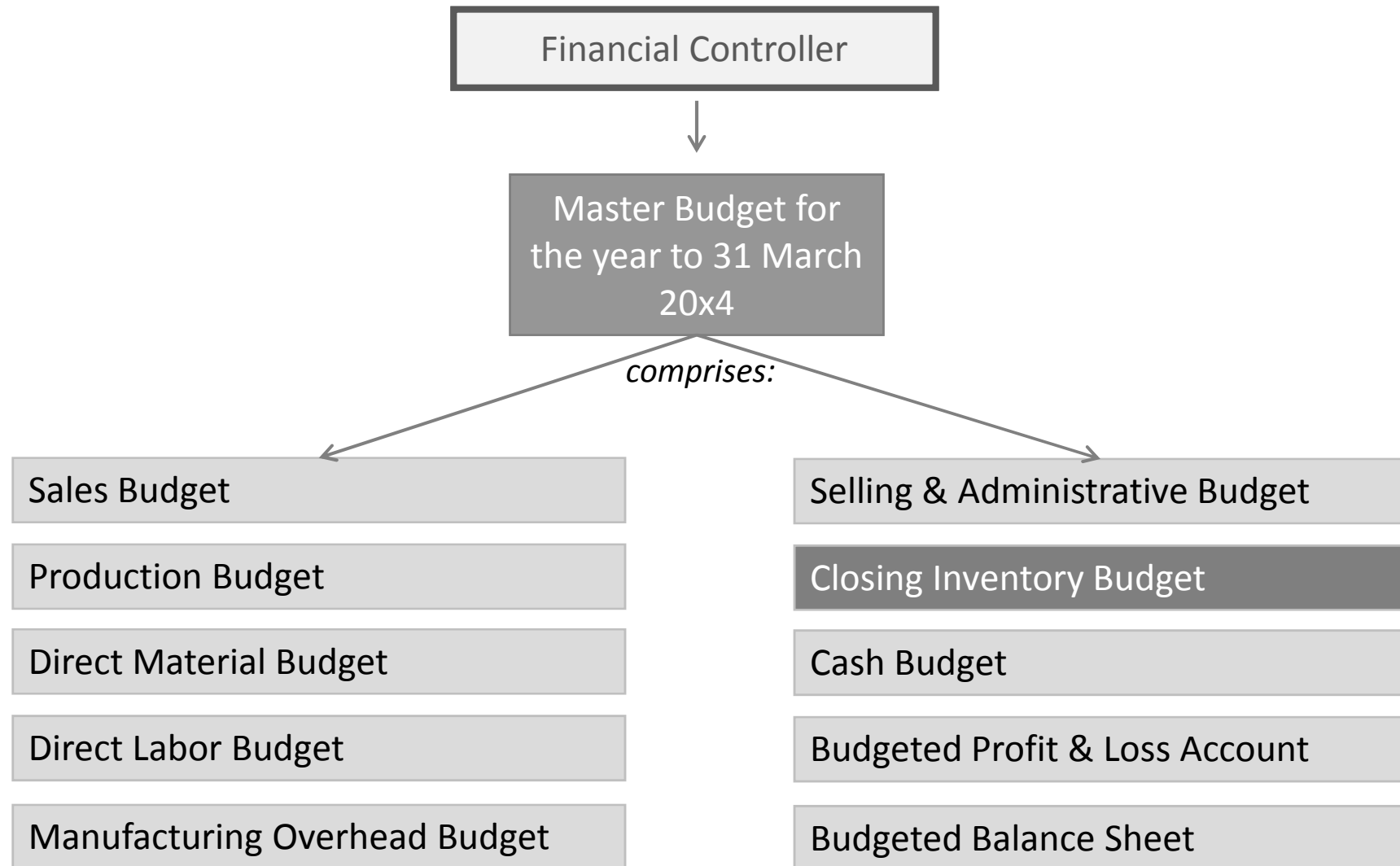
Management would expect a detailed analysis of variable and fixed overhead and would not accept manufacturing overhead budget without such an analysis

**12.4.6 Selling and Administrative Budget for Year Ending 31 March 20x4
(Schedule 6)**

	Q1	Q2	Q3	Q4	Total
Total sales	€175 000	€196 000	€119 000	€210 000	€700 000
Selling and admin. variable					
Advertising 5%	€8 750	€9 800	€5 950	€10 500	€35 000
R&D 3%	5 250	5 880	3 570	6 300	21 000
Fixed					
Salaries	2 000	2 000	2 000	2 000	8 000
Consumables	2 000	2 000	2 000	2 000	8 000
Advertising	3 000	3 000	3 000	3 000	12 000
Total selling and admin.	€21 000	€22 680	€16 520	€23 800	€84 000

The variable component of this budget for Advertising and R&D is based on the quarterly sales revenue figure as entered in the financial accounts (not as received in the form of cash)

Master Budget



Now the financial controller would pull the various components together in the form of a cash budget and budgeted profit and loss account and balance sheet

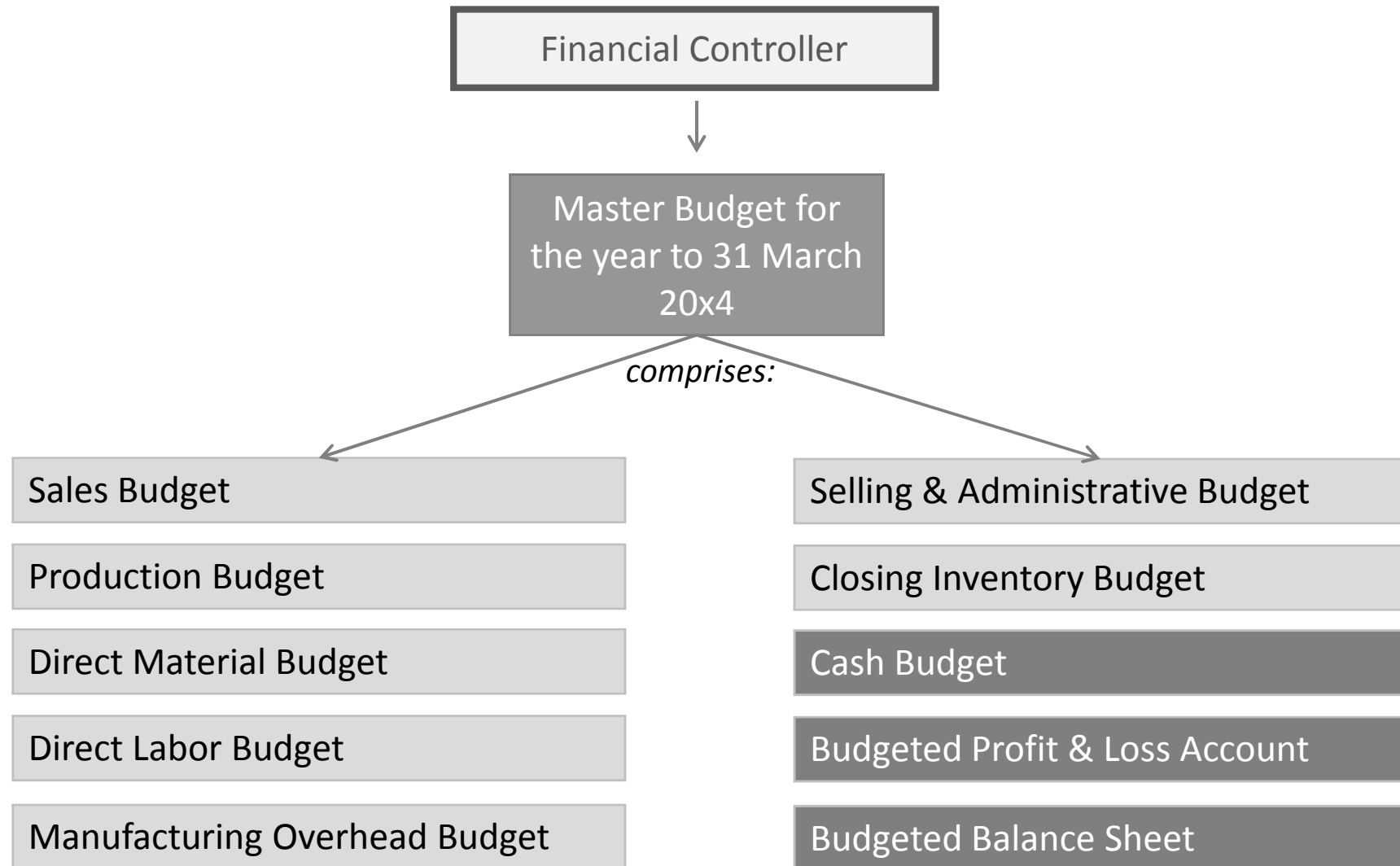
But first we'll monitor closing balances for inventory

	Q1	Q2	Q3	Q4	Total
Direct materials					
Units × €1 per unit	€2 580	€1 960	€3 000	€3 000	€3 000
Finished goods					
Units	560	340	600	600	600
× variable cost	× €55	× €55	× €55	× €55	× €55
	€30 800	€18 700	€33 000	€33 000	€33 000

The finished goods inventory valuation is, €55 each, that is the variable cost of manufacture

Businesses carry inventory in their balance sheets 'at the lower of cost or market value'; if the market value is less than cost the company would be required to write off the difference to the P\L account

Master Budget



The cash budget comprises those elements from previous schedules which have an impact on the inflows and outflows of cash over the next 12 months

12.4.8 Cash Budget for Year Ending 31 March 20x4 (Schedule 8)

	Q1	Q2	Q3	Q4	Total
Opening balance	€17 500	€(9 668)	€14 132	€28 156	€17 500
Cash from sales	130 000	189 700	142 100	182 700	644 500
Cash inflows	€147 500	€180 032	€156 232	€210 856	€662 000
Direct materials	€19 168	€26 220	€22 456	€26 256	€94 100
Direct labour	77 400	77 400	58 800	90 000	303 600
Manufacturing overhead	39 600	39 600	30 300	45 900	155 400
Selling and administrative	21 000	22 680	16 520	23 800	84 000
	€157 168	€165 900	€128 076	€185 956	€637 100
Closing balance	€(9 668)	€14 132	€28 156	€24 900	€24 900

*Closing cash balance of €17,500
in the balance sheet as at 31
March 20x3*

Q1 (Schedule 1(b))

Q1 (Schedule 6)

Q1 (Schedule 5)

Q1 (Schedule 4)

Q1 (Schedule 3(c))

*Q1 deficit of €9668, forms the
opening balance in Q2*

The managerial significance of the Cash Budget (Schedule 8) should not be ignored because cash is not an asset that works well for a business

Notes

Deficit
balance in Q1

Caused by a low debtors figure to collect from the end of 20x3; this would provoke the financial director to arrange bridging facilities with the company's bankers, the build-up of cash needs to be addressed

Cash

Although it earns interest, and it provides a cushion against which the company can fall in times of hardship, but it is not productive in the real meaning of the word.

Could the company contemplate a better strategic use of this amount? Investment in more R&D? Purchase of more capacity? Acquisition of another business – a competitor or supplier?

The construction of the budgeted P\L account and balance sheet follows the principles outlined in Modules 1 through 6

12.4.9 Budgeted Profit and Loss Account for Year Ending 31 March 20x4 (Schedule 9)

	Q1	Q2	Q3	Q4	Total
Sales units	2 500	2 800	1 700	3 000	10 000
	€	€	€	€	€
Sales revenue	175 000	196 000	119 000	210 000	700 000
Variable cost of sales manufacturing	(137 500)	(154 000)	(93 500)	(165 000)	(550 000)
Selling and administrative	(14 000)	(15 680)	(9 520)	(16 800)	(56 000)
Contribution margin	23 500	26 320	15 980	28 200	94 000
Fixed costs of manufacturing	(2 400)	(2 400)	(2 400)	(2 400)	(9 600)
Selling and administrative	(7 000)	(7 000)	(7 000)	(7 000)	(28 000)
Profit before tax	14 100	16 920	6 580	18 800	56 400

For simplicity we've ignored the impact of taxation; where usually it would be calculated for 20x3/x4 on the final budgeted profit of €56,400 but not paid until the following year

12.4.10 Budgeted Balance Sheet for Year Ending 31 March 20x4 (Schedule 10)

	Q1	Q2	Q3	Q4	Total
Plant and equipment	75 000	75 000	75 000	75 000	75 000
Accumulated depreciation	(26 500)	(28 000)	(29 500)	(31 000)	(31 000)
Inventory:					
Raw materials	2 580	1 960	3 000	3 000	3 000
Finished goods	30 800	18 700	33 000	33 000	33 000
Debtors	52 500	58 800	35 700	63 000	63 000
Cash	(9 668)	14 132	28 156	24 900	24 900
Total assets	124 712	140 592	145 356	167 900	167 900
Ordinary shares	30 000	30 000	30 000	30 000	30 000
Retained earnings	83 600	100 520	107 100	125 900	125 900
Creditors	11 112	10 072	8 256	12 000	12 000
Total equity and liabilities	124 712	140 592	145 356	167 900	167 900

Opening balance of €25,000 is added to at the rate of €1,500 per quarter

From Schedule 1(b); the sums of money collected in the next quarter are the debtors for the current quarter

From Schedule 3(c) we see that €11,112 paid in Q2 refers to purchases made in Q1

The opening figure for 20x3/x4 is €69,500 plus 14,100 added profit from the P\L Account

Agenda

12.1 Introduction

12.2 Why Bother with Budgets?

12.3 Why Budgeting Gets a Bad Name

12.4 Budgeting in Action: The Go-Straight Trolley Company

12.5 Discretionary Expenditure and Zero-Base Budgeting

Review Questions

Case Studies

Remember: Discretionary costs are those costs that can be influenced by managerial decision

Engineered Costs

Costs engineered into the product

Should be incurred every accounting period

e.g.: the cost of material and direct labor or certain critical items of manufacturing overhead

Discretionary Costs

Costs that managers can reduce relatively quickly

Need not be incurred every accounting period

e.g.: R&D, marketing, aspects of administration, machine maintenance and legal services

There are problems associated with these discretionary costs

First: discretionary costs are so difficult to budget and to manage; that's because of the difficulty surrounding the measurement of output

Example

In a **pharmaceutical company's R&D** lab a successful piece of research may culminate in the conclusion that a certain mixture of compounds would be harmful or ineffective; for the research scientist such an outcome is as valid as one which leads to the development of a new drug. It would therefore be wrong to measure output in terms only of successful experiments.

Example

In a **legal department**, the temptation to measure output in terms of lawsuits defended or activated must be resisted when one considers that perhaps the most successful legal work is done behind the scenes and is that which prevents any public action being taken.

Second: another problem is that the budget holders are not sure of the corporate objectives; and consequently unsure of where their efforts should be focused

Example

- In a **pharmaceutical company's R&D**, scientists tend to be subject to this breakdown in communication; if care is not taken, pockets of scientific expertise can beaver away in areas of investigative work which mirror individual interest rather than the strategic thrust of the company.
- Careful top-level scrutiny of the budget is essential to overcome this problem, although it is not easy for non-scientists to penetrate the technical description of proposed work.

The third major issue concerning discretionary expenditure is that it is not necessarily easy to control on an annual basis

Example

Some R&D programs run over several years and it is not possible to terminate these without major disruption and upset to the morale of the staff involved. A year is a short time in science.



*All these
lead to:*

R&D managers or directors of legal services hide behind the mystique of their disciplines to grab more and more scarce corporate resources that are wasted on needless work

One technique designed to counter this fear is zero-base budgeting, where activities are ranked in terms of importance

Zero-Base Budgeting (ZBB)

- Instead of adopting the normal incremental approach to annual budgeting management invites certain activities to bid for their resources as if they were starting from scratch (or from a zero base).
- ZBB involves the **analysis of activities** such as R&D or legal services or machine maintenance into packages of work which can be separated from each other, starting with the **most fundamental activity** and building up to the activity which, were the resources to be made available, it would be beneficial to undertake.
- These packages of discrete activities are costed and, when placed alongside packages from other discretionary activities, **ranked in order of top management priority**.
- It requires corporate functions to identify minimum levels of expenditure below which their activities cannot really operate; where new requests for increased spending are given a priority against existing commitments.

These decisions are extremely difficult but the secret of ZBB is that the decisions are made by top management and not by budget holders who would sure have a conflict of interest

Example

Consider the following packages which have been presented to the top management of an electronics company whose discretionary budget has been set at €250,000. (Read the table below from the bottom to the top):

Research and development		Legal services	
		Package 4 of 4. One lawyer to handle legal aspects surrounding intellectual property rights, copyrights and trademarks.	
		Further computer facilities.	€40 000
Package 3 of 3. Two software scientists to design and write customised operating systems; to work with electronic engineers in design of new products.	€70 000	Package 3 of 4. Two further lawyers to handle all contracts and property transactions.	
		Database purchase in support.	€80 000
Package 2 of 3. Two electronic engineers to design and build own prototypes.	€70 000	Package 2 of 4. Fully qualified lawyer to handle routine contractual problems. External consultant still to handle complex contracts.	€30 000
Package 1 of 3. Development engineer to test bought-in components and prototypes, altering the designs as required.	€40 000	Package 1 of 4. Partially qualified assistant to liaise with external consultant lawyers who undertake all legal work.	€10 000
	<u>€180 000</u>		<u>€160 000</u>

Example (Continued)

Management of this electronics company is faced with packages totaling €340,000 to be fitted into a budget of €250,000. On the basis that the budget limit is indeed the maximum amount, possible packages worth €90,000 will have to be dropped. A likely scenario may be as follows:

1. Each Package 1 would be accepted as representing the absolute minimum activity required to keep alive R&D and legal services. (It is possible, however, that management could decide to close down either function and allow the other function to go ahead with all packages.) Amount used €50,000.
2. Legal Package 2 provides a minimum level of in-house legal service which can be built on in future years. It also represents less than half the commitment of R&D Package. Equally it could be argued that a buoyant R&D activity represents the lifeblood of the company in the future; there is little point in keeping an in-house legal service to negotiate contracts if there are no new products to sell and no contracts to negotiate! Both Level 2 packages could be accepted; amounts used so far – €150,000.

Example (Continued)

3. The next layer of ranking becomes more difficult and depends on a top-level strategic view of the future of the company. Can a company afford to turn its back on technology at the expense of legal services? But can the company be assured of receiving value for money for the €70,000 to be spent on the proposed software initiative? If a positive decision is made for the R&D Level 3 package, can management be certain that no further support costs will emerge before the next budget round? A likely outcome could be agreement on R&D Level 3 (amount used so far, €220,000) and the two departments invited to resubmit packages totaling €30,000 each. Then the ranking process could adjudicate on these top-up packages.

The principal advantage of ZBB, as seen from the above example, is the involvement of top management in the allocation of discretionary funds

The principal advantage of ZBB is the involvement of top management in the allocation of discretionary funds. Too often senior executives who have generalist backgrounds fail to wrestle with expert fields such as R&D and law; in these circumstances, budgets tend to be 'noddled through' without any scrutiny.

ZBB forces top management to become involved and to relate the budget expectations of experts with the strategic direction of the business.

ZBB however suffers from two significant drawbacks: it involves extensive effort to rank activities and it is fairly subjective

1 The process of cutting up relatively homogeneous activities into discrete, separately costed packages is fraught with problems and involves considerable effort (and therefore cost) on the part of the staff of the functions involved. The sheer effort involved in implementing ZBB in some companies has been a barrier to extending the technique to further areas.

2 The definition of packages is left, inevitably, to the management of the function under review. The vested interest locked up in the process must leave some questions hanging over the packages and the order in which they are presented. The temptation with ZBB is to define the packages in such a way that the status quo is maintained. This can be achieved by submitting a fulsome Level 1 package. Top management may need to seek external expert opinion on the definitions if it is in doubt about the requests being made.

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Multiple Choice Questions (Continued)

12.1 The task of budget preparation produces clear benefits to companies. These include:

- i. increasing the motivation of individual heads of departments;
- ii. the provision of a benchmark against which to measure actual performance;
- iii. an explanation of the details in last year's profit and loss account;
- iv. improved co-ordination of activities between departments.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i), (ii) and (iii) only.
- C. (i), (ii) and (iv) only.
- D. (ii), (iii) and (iv) only.

Multiple Choice Questions (Continued)

12.2 The budgeting process is vital to a company because it is a financial statement which focuses on a comparison of historical costs. True or false?

False

Multiple Choice Questions (Continued)

12.3 Which of the following describes a budget?

- A. A strategic plan.
- B. A cash flow forecast.
- C. A review of management performance.
- D. A numerical expression of a plan of action.

Multiple Choice Questions (Continued)

12.4 If middle management has confidence in the achievability of the budget, then it is wholly irrelevant if senior management chooses to ignore the entire budget process. True or false?

False

Multiple Choice Questions (Continued)

12.5 Unfortunately, many members of management teams hold adverse views on the budgeting process. These opinions are misguided, but have to be tackled by taking steps to ensure that:

- i. managers in operating divisions are held responsible for costs and revenues over which they have no control;
- ii. any existing inefficiencies in budgets are driven out when resources are reduced;
- iii. the budget process begins as late as possible in the current year, compatible with obtaining full agreement before the beginning of the next year;
- iv. budgets restrict individuals and activities to the absolute minimum level of resources needed to perform an average job.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (ii) and (iii) only.
- C. (ii), (iii) and (iv) only.
- D. (iii) and (iv) only.

Multiple Choice Questions (Continued)

12.6 One of the problems of the budgeting process is that external circumstances can change significantly at an early stage in the year, with the result that the original budget figures become meaningless. The negative effects of such changes can be sensibly minimized if management:

- i. introduces a rolling budget system;
- ii. authorizes a general increase in budgeted costs without demanding an equivalent rise in budgeted sales;
- iii. eliminates the financial effect of such external changes to focus on variances arising where managers are clearly accountable;
- iv. permits surplus personnel to be transferred to other departments.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i) and (iii) only.
- C. (ii) and (iii) only.
- D. (iii) and (iv) only.

Multiple Choice Questions (Continued)

12.7 If a company operates in a market with unlimited demand for its product, which of the following is the first budget to be prepared?

- A. Production budget.
- B. Cash budget.
- C. Sales budget.
- D. Purchases budget.